OFFICIAL IDEA & SYSTEM ARCHITECTURE SPECIFICATION | 100% OPERATIONAL PROTOTYPE

SkyReady: AI/ML Predictive Maintenance & Fleet Availability Optimizer

Google OR-Tools CP-SAT Closed-Loop Scheduling, Quantile RUL Regression & Air-Gapped Cryptographic Compliance

Team: ZEN (KILLSWITCHX) **Squadron:** No. 45 Flying Daggers (24 Fighters) **Station:** AFS Sulur / Forward Base **Ref Slide Deck:** D:\SKY ROCKET.pptx

FLEET AVAILABILITY	READY AIRCRAFT-DAYS	SOLVER LATENCY	WORK BUNDLING
86.8%	625 / 720	82.0 ms	100%
+15.4 pt gain vs reactive (71.4%)	+111 days gained (30d window)	Google OR-Tools CP-SAT	Eliminates fragmented bay visits

1. Executive Thesis: The "One Loop" Paradigm

In military aviation, maintenance practices are frequently fragmented and reactive: engine telemetry logbooks, depot spare inventories, technician rosters, and hangar slot schedules operate in isolated silos. A component failure probability is completely inactionable if the spare part has a 14-day lead time or if no qualified technician is rostered.

SkyReady breaks these disjoint silos through an integrated, closed-loop pipeline:

CLOSED-LOOP AIR COMBAT READINESS PIPELINE (SENSOR → SCHEDULE → SPARES)

01	02	03	04	05	06
DETECT	**PREDICT**	**EXPLAIN**	**OPTIMIZE**	**SIMULATE**	**ACT**
Isolation Forest on multi-variate C-MAPSS telemetry	Quantile RUL Regression [P10, P50, P90] safe bounds	SHAP sensor attribution & plain-language records	Google OR-Tools CP-SAT maximizes ready aircraft-days	Dynamic What-If sandbox (surge, lag, bay outages)	Officer release sign-off & SHA-256 tamper ledger

2. Technical Architecture & System Layers

System Layer	Core Technologies & Engines	Operational Function
1. Ingestion Layer	`NASA C-MAPSS FD001` Synthetic Fleet Generator (24 airframes)	Ingests 7 critical turbofan sensor channels (EGT, HPC pressure, core shaft RPM, rotor vibration, BPR).
2. ML Analytics	`Scikit-Learn` Isolation Forest & Quantile Gradient Boosting	Detects acute thermodynamic anomalies; calculates 90% confidence RUL intervals (P10 to P90).
3. Explainability	`TreeSHAP` Normalized Feature Deviation Engine	Generates plain-language aeronautical engineering narratives for flight-line crew & defense officers.
4. Optimization Core	`Google OR-Tools CP-SAT` Constraint Programming	Schedules 24 aircraft over 30 days across 3 hangar bays, 12 technician rosters, and spare lead times.
5. Stress Simulator	Dynamic Sensitivity Simulation Engine	Evaluates supply delays (+0 to +14d), combat surges (1.0x to 1.8x), and bay closures in <100ms.
6. Cryptographic Ledger	`SHA-256` Hash-Chained Blockchain Ledger	Air-gapped tamper-evident audit logging for officer release authorizations and manual overrides.

System Layer	Core Technologies & Engines	Operational Function
7. Web & REST API	`FastAPI` Uvicorn, Chart.js, HTML5 Tactical C2	Sub-10ms RESTful endpoints and real-time interactive military web cockpit dashboard.

3. Mathematical Formulation of the CP-SAT Optimizer

The optimizer schedules aircraft maintenance tasks across a 30-day horizon ($H = 30$). For each aircraft $i \in \{1, \dots, 24\}$, let S_i be the maintenance start day, d_i the duration, $E_i = S_i + d_i$ the end day, and $b_{i,j} \in \{0, 1\}$ the assignment to hangar bay $j \in \{\text{Bay-1, Bay-2, Bay-3}\}$.

Objective Function (Maximize Total Ready Aircraft-Days):
$$\max \sum_{i=1}^{24} \left(25 \cdot \text{IsScheduled}_i - 12 \cdot \text{IsLate}_i - 1 \cdot S_i \right)$$ **Subject to Hard Operational Constraints:**

1. *Pessimistic RUL Safety Deadline:* $S_i \le \text{SafeDays}_i - d_i$ (enforced with late penalty L_i if violated)
2. *Spares Lead-Time Arrival:* $S_i \ge \text{SpareLeadTime}_i + \text{SupplyLag}$
3. *Hangar Bay No-Overlap:* $\text{AddNoOverlap}\left(\{ [S_i, E_i] \mid b_{i,j} = 1 \}\right) \quad \forall j \in \{\text{Bay 1, 2, 3}\}$
4. *Technician Roster Cumulative Limits:* $\text{AddCumulative}\left(\{ [S_i, E_i, \text{Techs}_i] \}, \text{DailyTechCap}\right)$
5. *Work Bundling:* Consolidated single bay entry per airframe visit.

4. Comparative Operational Benchmarks

Evaluation Dimension	Reactive Maintenance	Calendar-Based Schedule	SkyReady (CP-SAT)	Operational Impact
Fleet Availability Rate	71.4%	76.2%	**86.8%**	`+15.4 pt gain` (target +5-10 pt)
30-Day Ready Aircraft-Days	514 days	548 days	**625 days**	`+111 Ready Fighter-Days`
Unplanned Groundings	18 events / year	12 events / year	**4 events / year**	`~40% to 70% Reduction`
Planning Re-Solve Time	6 to 8 hours (manual)	2 hours	**82.0 milliseconds**	Real-time tactical agility
Audit & Compliance	Paper logbooks	Database rows	**SHA-256 Chained**	Air-gapped tamper-evident audit

5. REST API Endpoints & Automated Verification

Production REST API Endpoints

Method & Route	Purpose
`GET /`	Tactical C2 Single-Page Web Dashboard
`GET /api/fleet`	24-Aircraft status, health & KPIs
`GET /api/fleet/{tail}`	C-MAPSS telemetry & SHAP breakdown
`GET /api/optimizer/schedule`	30-Day CP-SAT Gantt & Requisitions
`POST /api/simulator/what-if`	Dynamic sensitivity sandbox with sliders
`GET /api/audit/verify`	Validates cryptographic hash chain

Automated Test Suite (100% Pass Rate)

✓ 8/8 Pytest Verification Tests Passing (100%)

- test_fleet_generation: PASSED
- test_anomaly_detection_healthy_vs_degraded: PASSED
- test_rul_quantile_estimator: PASSED
- test_shap_explainability_and_narrative: PASSED
- test_cpsat_fleet_optimizer_execution: PASSED
- test_what_if_stress_simulation: PASSED
- test_sha256_cryptographic_audit_ledger: PASSED
- test_api_rest_endpoints: PASSED

One-Click Startup: `start.bat` | `start.ps1` | `start.sh`
Web Access: http://localhost:8000 | Docs: /docs